

The empirical recurrence for OEIS A301487, recovered from its tail

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Abstract

OEIS A301487 records “Empirical recurrence of order 89” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 110$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{89} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-89 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A301487 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 1, 2, 4, 5 or 6 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

8, 50, 148, 599, 2354, 9415, 38558, 158994, ...

The entry states:

Empirical recurrence of order 89 (see link above)

As of the “Last modified” line on the live entry (Mar 22 2018 11:46:56, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 110$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 110$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 89: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 89.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 220$ exact terms from $n = 3$ on, it returns order 89 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{89} c_i a(n-i),$$

c_1	15	c_{24}	−156664	c_{47}	306358606	c_{70}	−579208
c_2	−95	c_{25}	531172	c_{48}	−113299080	c_{71}	−933402
c_3	344	c_{26}	296179	c_{49}	−342305439	c_{72}	−614103
c_4	−863	c_{27}	1978794	c_{50}	−337386707	c_{73}	−191502
c_5	1861	c_{28}	−1489733	c_{51}	−194629481	c_{74}	43745
c_6	−3798	c_{29}	−2977236	c_{52}	−8660492	c_{75}	135692
c_7	6604	c_{30}	−7193002	c_{53}	103466234	c_{76}	95535
c_8	−9937	c_{31}	−6879394	c_{54}	137156048	c_{77}	32405
c_9	16199	c_{32}	12747983	c_{55}	151839244	c_{78}	−2974
c_{10}	−26368	c_{33}	27280911	c_{56}	131133624	c_{79}	−8100
c_{11}	34323	c_{34}	14323609	c_{57}	32059734	c_{80}	−4468
c_{12}	−43777	c_{35}	−23026267	c_{58}	−82156653	c_{81}	−1627
c_{13}	64032	c_{36}	−58790360	c_{59}	−100561807	c_{82}	621
c_{14}	−82326	c_{37}	−44287430	c_{60}	−42030869	c_{83}	310
c_{15}	115572	c_{38}	62602795	c_{61}	−2153963	c_{84}	−30
c_{16}	−164864	c_{39}	145948129	c_{62}	2422783	c_{85}	−8
c_{17}	166985	c_{40}	91977952	c_{63}	3135741	c_{86}	−37
c_{18}	−230196	c_{41}	−66510239	c_{64}	9747633	c_{87}	−6
c_{19}	294272	c_{42}	−255751099	c_{65}	11912174	c_{88}	3
c_{20}	−207998	c_{43}	−313005526	c_{66}	5042770	c_{89}	1
c_{21}	429318	c_{44}	−87917734	c_{67}	−418363		
c_{22}	−747082	c_{45}	313919988	c_{68}	−575601		
c_{23}	347972	c_{46}	528391143	c_{69}	−113445		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A301487 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{89} - c_1x^{89-1} - \dots - c_{89}$. Its constant term is $-c_{89} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 89 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 89, so $\deg q = 89$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 89, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 89, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-89 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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